Smart Medical Records Platform: Class Specification

Graduation Project · Healthcare · Web-Based MVP · Version 1.0

Derived from: Business Requirements Document (BRD), Software Requirements Specification (SRS), and the Final ERD.

1. Overview

This document specifies the domain classes of the Smart Medical Records Platform. The system is doctor-centric: the **Doctor** is the authenticated user and the **Patient** is a medical record entity with no account or login. Every medical event (visit, diagnosis, medication, lab result, document, note) belongs to a specific patient.

Conventions

- **PK** = primary key, **FK** = foreign key, **Req** = required field.
- Types: `UUID`, `String`, `Text`, `Date`, `DateTime`, `Time`, `Decimal`, `Int`, `Enum`.
- Methods are listed as public operations of the domain layer; each maps to SRS functional requirements (FR).

2. Class Summary

#	Class	Responsibility	Main SRS Reference
1	Doctor	Authenticated user who owns patient records	FR-01 to FR-07
2	Patient	Medical record entity (not a user)	FR-08 to FR-13
3	MedicalVisit	A consultation between a doctor and a patient	FR-14 to FR-16
4	Diagnosis	Diagnosis recorded during a visit	FR-17, FR-18
5	Medication	Prescribed medication (prescription item)	FR-19 to FR-21
6	Allergy	Known patient allergy	FR-22, FR-23
7	ChronicCondition	Long-term patient condition	FR-24, FR-25
8	VitalSigns	Measurements taken during a visit	FR-26, FR-27
9	LaboratoryResult	Lab test result	FR-28, FR-29

#	Class	Responsibility	Main SRS Reference
10	MedicalNote	Clinical note on a patient or visit	FR-30, FR-31
11	MedicalDocument	Uploaded file (PDF/ JPG/JPEG/PNG)	FR-32 to FR-35
12	Secretary	Authenticated user who manages appointments (not a medical user)	FR-42, FR-49, FR-50
13	Appointment	Scheduled booking between a patient and a doctor	FR-43 to FR-48, FR-51

3. Class Relationships

From	Relationship	To	Cardinality
Doctor	manages	Patient	1 : M
Doctor	performs	MedicalVisit	1 : M
Doctor	creates	MedicalNote	1 : M
Doctor	uploads	MedicalDocument	1 : M
Patient	has	MedicalVisit	1 : M
Patient	has	Allergy	1 : M
Patient	has	ChronicCondition	1 : M
Patient	has	MedicalDocument	1 : M
Patient	undergoes	LaboratoryResult	1 : M
MedicalVisit	records	VitalSigns	1 : M
MedicalVisit	includes	Diagnosis	1 : M
MedicalVisit	prescribes	Medication	1 : M
MedicalVisit	generates	MedicalNote	1 : M
Secretary	assists	Doctor	M : M
Secretary	creates	Appointment	1 : M
Doctor	receives	Appointment	1 : M
Patient	books	Appointment	1 : M
Appointment	leads to	MedicalVisit	1 : 0..1

4. Class Specifications

4.1 Doctor

Purpose: The only authenticated user of the platform; owns a workspace of patients.

Attribute	Type	Key	Notes
doctorId	UUID	PK	Auto-generated
fullName	String	Req	
email	String	Req	Unique, valid format
passwordHash	String	Req	Never stored in plain text
phone	String		
specialization	String		
accountStatus	Enum	Req	Active / Inactive
createdAt	DateTime	Req	

Methods: `register()`, `login()`, `logout()`, `resetPassword()`, `viewProfile()`, `updateProfile()`, `viewAppointments()`

Rules: A doctor can only access patients in their own workspace (FR-13, BRULE-04).

4.2 Patient

Purpose: Central medical record entity. Patients cannot register or log in.

Attribute	Type	Key	Notes
patientId	UUID	PK	
doctorId	UUID	FK	Owning doctor
fullName	String	Req	
dateOfBirth	Date	Req	Age is derived (see Section 5)
gender	Enum	Req	
phone	String		Searchable
address	String		
bloodType	Enum		A+, A-, B+, B-, AB+, AB-, O+, O-
emergencyContact	String		
createdAt	DateTime	Req	

Methods: `create(), update(), search(name | id | phone), viewProfile(), getTimeline()`

Rules: Every medical event must be linked to a patient (BRULE-05). A secretary may search patients and create/update basic information only (FR-49), never medical data (BRULE-10).

4.3 MedicalVisit

Purpose: One consultation; the parent of diagnoses, medications, vital signs, and notes.

Attribute	Type	Key	Notes
visitId	UUID	PK	
patientId	UUID	FK	Req
doctorId	UUID	FK	Req
appointmentId	UUID	FK	Optional link to an appointment
visitDate	DateTime	Req	
chiefComplaint	Text		
symptoms	Text		
clinicalNotes	Text		

Methods: `create(), update(), getPreviousVisits()`

Rules: A visit must belong to one patient and one doctor (BRULE-06).

4.4 Diagnosis

Attribute	Type	Key	Notes
diagnosisId	UUID	PK	
visitId	UUID	FK	Req
patientId	UUID	FK	Req
diagnosisName	String	Req	
description	Text		
diagnosisDate	Date	Req	

Methods: `add(), getHistoryByPatient()`

4.5 Medication

Purpose: A prescribed medication; a visit's prescription is its collection of medications.

Attribute	Type	Key	Notes
medicationId	UUID	PK	

Attribute	Type	Key	Notes
visitId	UUID	FK	Req
patientId	UUID	FK	Req
medicationName	String	Req	
dosage	String	Req	
frequency	String	Req	
duration	String		
instructions	Text		

Methods: `add(), getHistory(), createPrescription(visitId)`

4.6 Allergy

Attribute	Type	Key	Notes
allergyId	UUID	PK	
patientId	UUID	FK	Req
allergyName	String	Req	
severity	Enum		Mild / Moderate / Severe
description	Text		
notes	Text		

Methods: `record(), listByPatient()`

Rules: Allergies are displayed prominently in the patient profile (FR-23).

4.7 ChronicCondition

Attribute	Type	Key	Notes
conditionId	UUID	PK	
patientId	UUID	FK	Req
conditionName	String	Req	e.g. Hypertension, Diabetes
diagnosisDate	Date		
status	Enum		Active / Controlled / Resolved
notes	Text		

Methods: `record(), listByPatient()`

Rules: The system never diagnoses automatically (FR-24).

4.8 VitalSigns

Attribute	Type	Key	Notes
vitalSignsId	UUID	PK	
visitId	UUID	FK	Req
bloodPressure	String		e.g. 120/80
heartRate	Int		bpm
temperature	Decimal		°C
respiratoryRate	Int		breaths/min
weight	Decimal		kg
height	Decimal		cm
spO2	Int		%

Methods: `record(), getHistory()`

4.9 LaboratoryResult

Attribute	Type	Key	Notes
resultId	UUID	PK	
patientId	UUID	FK	Req
visitId	UUID	FK	Optional link to a visit
testName	String	Req	
testDate	Date	Req	
value	Decimal	Req	Numeric validation
unit	String	Req	
referenceRange	String		
notes	Text		

Methods: `add(), getHistory()`

4.10 MedicalNote

Attribute	Type	Key	Notes
noteId	UUID	PK	
patientId	UUID	FK	Req

Attribute	Type	Key	Notes
doctorId	UUID	FK	Req
visitId	UUID	FK	Optional
noteContent	Text	Req	
createdDate	DateTime	Req	
updatedDate	DateTime		

Methods: `add(), update(), listByPatient()`

4.11 MedicalDocument

Attribute	Type	Key	Notes
documentId	UUID	PK	
patientId	UUID	FK	Req
uploadedBy	UUID	FK	Doctor who uploaded
fileName	String	Req	
type	Enum	Req	PDF / JPG / JPEG / PNG
location	String	Req	Storage path or URL
uploadDate	DateTime	Req	
description	Text		

Methods: `upload(), view(), download()`

Rules: Validate file format and size limits before storing (SRS Section 11).

4.12 Secretary

Purpose: Authenticated user who manages appointments for assigned doctors; not a medical user.

Attribute	Type	Key	Notes
secretaryId	UUID	PK	Auto-generated
fullName	String	Req	
email	String	Req	Unique, valid format
passwordHash	String	Req	Never stored in plain text
phone	String		
accountStatus	Enum	Req	Active / Inactive

Attribute	Type	Key	Notes
createdAt	DateTime	Req	

Methods: `login(), logout(), resetPassword(), searchPatient(), createAppointment(), rescheduleAppointment(), cancelAppointment(), viewSchedule()`

Rules: A secretary can only access appointments and basic patient information of the doctor(s) assigned to them, and never medical records (FR-50, BRULE-10). Doctor assignment is a many-to-many link (Secretary assists Doctor).

4.13 Appointment

Purpose: A scheduled booking between a patient and a doctor on a specific date and time.

Attribute	Type	Key	Notes
appointmentId	UUID	PK	
patientId	UUID	FK	Req
doctorId	UUID	FK	Req
createdBy	UUID	FK	Req; secretary or doctor who booked
appointmentDate	Date	Req	
startTime	Time	Req	
endTime	Time		
reason	Text		
status	Enum	Req	Scheduled / Confirmed / Completed / Cancelled / No-Show
notes	Text		
cancellationReason	Text		
createdAt	DateTime	Req	

Methods: `create(), reschedule(), cancel(), updateStatus(), listByDoctor(), listByPatient()`

Rules: An appointment must belong to one patient and one doctor, and cannot overlap another appointment of the same doctor (FR-48, BRULE-11).

5. Business Rules Summary

Rule	Description
BRULE-01	Only authenticated doctors can access records

Rule	Description
BRULE-02	Patients have no accounts or credentials
BRULE-03	A patient record belongs to its creating doctor's workspace
BRULE-04	A doctor sees only authorized patient records
BRULE-05	Every medical event belongs to a specific patient
BRULE-06	Every visit belongs to a patient and a doctor
BRULE-07	Every document belongs to a specific patient
BRULE-10	A secretary accesses only appointments and basic patient information of assigned doctors, never medical records
BRULE-11	Every appointment belongs to a patient and a doctor and cannot overlap another appointment of the same doctor